

A Rectangular-Band Counterexample Separating $WLUC(S)$ from $LMC(S)$

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Status

Counterexample, likely valid. This answers affirmatively the question in Wisnicki's *Amenable semigroups and nonexpansive dynamical systems* [1]: there is a Hausdorff topological semigroup S for which

$$WLUC(S) \subsetneq LMC(S).$$

Source Question

The source defines $f \in WLUC(S)$ when the orbit map

$$S \ni s \longmapsto L_s f \in C(S)$$

is continuous for the weak topology of $C(S)$, and $f \in LMC(S)$ when the same map is continuous for the topology induced by multiplicative means on $C(S)$. Since multiplicative means are among the bounded linear functionals, one always has $WLUC(S) \subseteq LMC(S)$. After Theorem 3.5, the source asks whether this inclusion can be proper.

8

A. WIŚNICKI

and $y \in K$. Now the result follows from Theorem 3.3 specialized to $Y = LMC(S)$. $\square$

It appears that $LMC(S)$ should be replaced by $WLUC(S)$ in Theorem 3.5 as in the case of affine actions (see [22, Theorem 4]) but it is out of our reach for now. In fact, it seems to be an open problem whether $WLUC(S)$ is a proper subspace of $LMC(S)$.

Idea of the Proof

Weak continuity tests a left-translation orbit against all bounded linear functionals on $C(S)$, while LMC -continuity tests only multiplicative means. The construction makes this gap visible. We build a rectangular band $S = A \times \mathbb{N}$, with multiplication $(a, m)(b, n) = (a, n)$. Left translates of one function f then depend only on the a -coordinate and on the $\mathbb{N}$ -coordinate.

The space A is arranged so that a net of finite subsets $E \subset \beta\mathbb{N}$ converges to a point ∞ . For each E , choose a continuous function u_E on $\beta\mathbb{N}$ which vanishes on E but still has large integral against a

fixed atomless probability measure. Vanishing at each prescribed ultrafilter gives continuity against every multiplicative mean. Large integrals give a single bounded linear functional witnessing failure of weak continuity.

The Counterexample

Counterexample. *There is a Hausdorff topological semigroup S and a function $f \in C_b(S)$ such that $f \in \text{LMC}(S)$ but $f \notin \text{WLUC}(S)$. Hence $\text{WLUC}(S)$ is a proper subspace of $\text{LMC}(S)$ in general.*

Proof. Let $\mathbb{N}$ have the discrete topology and let $\beta\mathbb{N}$ be its Stone-Cech compactification. First choose an atomless regular Borel probability measure μ on $\beta\mathbb{N}$. For example, take a map from $\mathbb{N}$ to $[0, 1]$ with dense range and extend it to a continuous surjection $q : \beta\mathbb{N} \rightarrow [0, 1]$. The Lebesgue state on $q^*C([0, 1]) \subset C(\beta\mathbb{N})$ extends to a state on $C(\beta\mathbb{N})$; by the Riesz representation theorem this state is integration against a regular probability measure μ , and $q_*\mu$ is Lebesgue measure. If μ had an atom at $p \in \beta\mathbb{N}$, then $q_*\mu$ would have an atom at $q(p)$, so μ is atomless.

For every finite set $E \in [\beta\mathbb{N}]^{<\omega}$, regularity gives an open set $U_E \supset E$ with $\mu(U_E) < 1/2$. By Urysohn's lemma choose

$$u_E \in C(\beta\mathbb{N}, [0, 1])$$

such that $u_E = 0$ on E and $u_E = 1$ on $\beta\mathbb{N} \setminus U_E$. Then

$$\int_{\beta\mathbb{N}} u_E d\mu \geq 1 - \mu(U_E) > 1/2. \quad (1)$$

Let

$$A = \{\infty\} \cup [\beta\mathbb{N}]^{<\omega}.$$

Every point $E \in [\beta\mathbb{N}]^{<\omega}$ is isolated, and a neighborhood basis of ∞ is

$$N_F = \{\infty\} \cup \{E \in [\beta\mathbb{N}]^{<\omega} : F \subseteq E\}, \quad F \in [\beta\mathbb{N}]^{<\omega}.$$

This space is Hausdorff: finite subsets are isolated, and ∞ is separated from $E_0 \in [\beta\mathbb{N}]^{<\omega}$ by $N_{\{p\}}$ for any $p \in \beta\mathbb{N} \setminus E_0$.

Now put

$$S = A \times \mathbb{N}$$

with the product topology and define multiplication by

$$(a, m)(b, n) = (a, n).$$

This is an associative rectangular-band multiplication, and it is jointly continuous because it is just the product of the first-coordinate projection from the left factor and the second-coordinate projection from the right factor. Thus S is a Hausdorff topological semigroup.

Define $f : S \rightarrow \mathbb{C}$ by

$$f(\infty, n) = 0, \quad f(E, n) = u_E(n) \quad (E \in [\beta\mathbb{N}]^{<\omega}, n \in \mathbb{N}),$$

where n is identified with its principal ultrafilter in $\beta\mathbb{N}$. The function is bounded by 1. It is continuous at isolated points (E, n) . At (∞, n) , the neighborhood $N_{\{n\}} \times \{n\}$ is sent to 0, since $E \supset \{n\}$ implies $u_E(n) = 0$. Hence $f \in C_b(S)$.

For $s = (a, m) \in S$, the left translate of f is

$$L_s f(b, n) = f((a, m)(b, n)) = f(a, n).$$

Thus $L_s f$ depends only on a and n . Let

$$j : \ell^\infty(\mathbb{N}) \longrightarrow C_b(S), \quad j(g)(a, n) = g(n).$$

This is an isometric unital $*$ -homomorphism. If $a = E \in [\beta\mathbb{N}]^{<\omega}$, then

$$L_{(E, m)} f = j(u_E|_{\mathbb{N}}),$$

and $L_{(\infty, m)} f = 0$.

We claim first that $f \in \text{LMC}(S)$. Let χ be any multiplicative mean on $C_b(S)$. The restriction $\chi \circ j$ is a character on $\ell^\infty(\mathbb{N}) = C(\beta\mathbb{N})$, so there is a point $p_\chi \in \beta\mathbb{N}$ such that

$$\chi(j(g)) = \widehat{g}(p_\chi) \quad (g \in \ell^\infty(\mathbb{N})).$$

Consequently,

$$\chi(L_{(E, m)} f) = u_E(p_\chi), \quad \chi(L_{(\infty, m)} f) = 0.$$

This scalar map is continuous at every isolated $E \in [\beta\mathbb{N}]^{<\omega}$. At ∞ , the neighborhood $N_{\{p_\chi\}} \times \mathbb{N}$ is mapped to 0, because every $E \supset \{p_\chi\}$ satisfies $u_E(p_\chi) = 0$. Hence

$$s \longmapsto \chi(L_s f)$$

is continuous for every multiplicative mean χ , exactly saying $f \in \text{LMC}(S)$.

It remains to show that $f \notin \text{WLUC}(S)$. On the closed subspace $j(\ell^\infty(\mathbb{N})) \subset C_b(S)$, define

$$M(j(g)) = \int_{\beta\mathbb{N}} \widehat{g} d\mu.$$

This is a bounded linear functional of norm 1, so by Hahn-Banach it has an extension $\Phi \in C_b(S)^*$. Direct $[\beta\mathbb{N}]^{<\omega}$ by inclusion and consider the net

$$s_E = (E, 0) \longrightarrow s_\infty = (\infty, 0)$$

in S . By (1),

$$\Phi(L_{s_E} f) = \int_{\beta\mathbb{N}} u_E d\mu > 1/2$$

for every $E \in [\beta\mathbb{N}]^{<\omega}$, while

$$\Phi(L_{s_\infty} f) = 0.$$

Therefore the orbit map $s \mapsto L_s f$ is not weakly continuous at s_∞ . Thus $f \notin \text{WLUC}(S)$.

We have exhibited $f \in \text{LMC}(S) \setminus \text{WLUC}(S)$, so the inclusion $\text{WLUC}(S) \subseteq \text{LMC}(S)$ is proper for this Hausdorff topological semigroup S . $\square$

Verification Notes

No numerical or finite search is used. The proof verifies the required properties directly:

- $S = A \times \mathbb{N}$ is a jointly continuous rectangular band.
- The constructed f is bounded and continuous because $\mathbb{N}$ is discrete and the neighborhoods $N_{\{n\}}$ force $u_E(n) = 0$ near ∞ .
- Every multiplicative mean on $C_b(S)$, restricted to functions depending only on the $\mathbb{N}$ -coordinate, is evaluation at an ultrafilter $p \in \beta\mathbb{N}$; the neighborhoods $N_{\{p\}}$ make the scalar orbit continuous.
- A Hahn-Banach extension of integration against μ witnesses failure of weak continuity.

Novelty Check and Limitations

Local run indexes searched on 2026-07-19:

`registry_index.tsv`, `solutions/index.tsv`, `attempts/index.tsv`, `proof_gaps/index.tsv`.

Search terms included arXiv:1909.09723, WLUC, LMC, “weakly left uniformly continuous”, “left multiplicatively continuous”, and “proper subspace”. No prior packet for this question was found.

External searches on 2026-07-19 used exact phrases including “WLUC(S) LMC(S) proper subspace”, “weakly left uniformly continuous left multiplicatively continuous”, and “RMC(S) WRUC(S) proper”. These found the source arXiv paper, the 2026 publication record for the same work, and standard background on semigroup compactifications, but no later explicit answer to this proper-subspace question.

The example is intentionally nonmetrizable and uses the Stone-Cech compactification of a discrete countable space. It answers the source question as stated for Hausdorff semitopological semigroups, and in fact for a topological semigroup. It does not decide whether equality may hold under additional hypotheses such as metrizability, local compactness of S , or first countability.

Human Review Recommendation

Accept as a likely valid counterexample to equality $WLUC(S) = LMC(S)$. Reviewers should focus on two points: the topology on A , especially the use of finite subsets of all of $\beta\mathbb{N}$, and the passage from multiplicative means on $C_b(S)$ to point evaluations on the $\mathbb{N}$ -coordinate subalgebra.

References

- [1] A. Wisnicki, *Amenable semigroups and nonexpansive dynamical systems*, arXiv:1909.09723v1, 2019.
- [2] L. Gillman and M. Jerison, *Rings of Continuous Functions*, Van Nostrand, 1960.
- [3] W. Rudin, *Functional Analysis*, 2nd ed., McGraw-Hill, 1991.